

EE111 Electric Circuits

Discussion 12

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Contents

- Frequency response
- Filters
- Passive Filters
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传递函数可以写为：

$$\mathbf{H} = H \angle \phi = H e^{j\phi} \quad (14.12)$$

两边取自然对数可以得到：

$$\ln \mathbf{H} = \ln H + \ln e^{j\phi} = \ln H + j\phi \quad (14.13)$$

因此， $\ln \mathbf{H}$ 的实部是幅度的函数，而其虚部就是相位。在幅度伯德图中，增益为：

$$\boxed{H_{\text{dB}} = 20 \log_{10} H} \quad (14.14)$$

$$\mathbf{H}(\omega) = \frac{K(j\omega)^{\pm 1} (1 + j\omega/z_1) [1 + j2\zeta_1 \omega/\omega_k + (j\omega/\omega_k)^2] \cdots}{(1 + j\omega/p_1) [1 + j2\zeta_2 \omega/\omega_n + (j\omega/\omega_n)^2] \cdots} \quad (14.15)$$

上式可以通过分配 $\mathbf{H}(\omega)$ 中的极点与零点而得到。式(14.15)所示的 $\mathbf{H}(\omega)$ 的表达式称为标准形式(standard form)。 $\mathbf{H}(\omega)$ 中可以包含多达七种不同的因子，这些因子可以是传递函数中各种不同的组合，它们是：

1. 增益 K ；
2. 在原点的极点 $(j\omega)^{-1}$ 或零点 $j\omega$ ；
3. 单极点 $1/(1 + j\omega/p_1)$ 或单零点 $(1 + j\omega/z_1)$ ；
4. 二阶极点 $1/[1 + j2\zeta_2 \omega/\omega_n + (j\omega/\omega_n)^2]$ 或二阶零点 $[1 + j2\zeta_1 \omega/\omega_k + (j\omega/\omega_k)^2]$ 。

在绘制伯德图时，首先分别绘制各因子的曲线，之后再将其相加起来。由于采用了对数运算，所以各因子可以单独考虑，再将它们相加组合成伯德图。正是因为对数在数学上便于处理，使得伯德图成为一种强有力的工程工具。

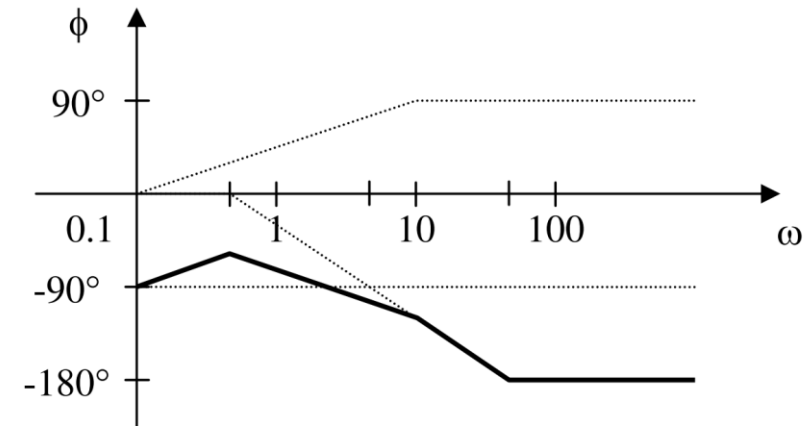
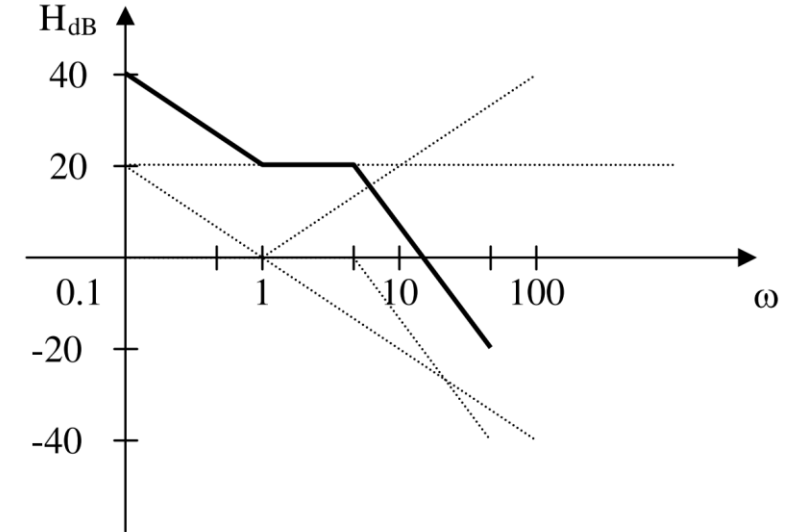
1. Draw the Bode plots for

$$\mathbf{H}(\omega) = \frac{250(j\omega + 1)}{j\omega(-\omega^2 + 10j\omega + 25)}$$

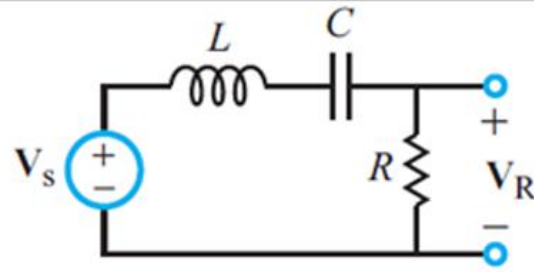
$$\mathbf{H}(\omega) = \frac{250}{25} \frac{1 + j\omega}{j\omega \left(1 + \frac{j\omega 10}{25} + \left(\frac{j\omega}{5} \right)^2 \right)}$$

$$H_{dB} = 20\log_{10} 10 + 20\log_{10} |1 + j\omega| - 20\log_{10} |j\omega| \\ - 20\log_{10} \left| 1 + j\omega 2/5 + (j\omega/5)^2 \right|$$

$$\phi = -90^\circ + \tan^{-1} \omega - \tan^{-1} \left(\frac{\omega 10/25}{1 - \omega^2/5} \right)$$



RLC Circuit



Transfer Function

$$H = \frac{V_R}{V_s}$$

Resonant Frequency, ω_0

$$\frac{1}{\sqrt{LC}}$$

Bandwidth, B

$$\frac{R}{L}$$

Quality Factor, Q

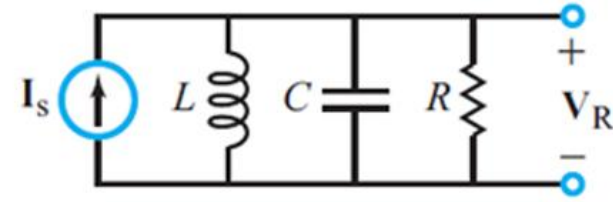
$$\frac{\omega_0}{B} = \frac{\omega_0 L}{R}$$

Lower Half-Power Frequency, ω_1

$$\left[-\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$

Upper Half-Power Frequency, ω_2

$$\left[\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$



$$H = \frac{V_R}{I_s}$$

$$\frac{1}{\sqrt{LC}}$$

$$\frac{1}{RC}$$

$$\frac{\omega_0}{B} = \frac{R}{\omega_0 L}$$

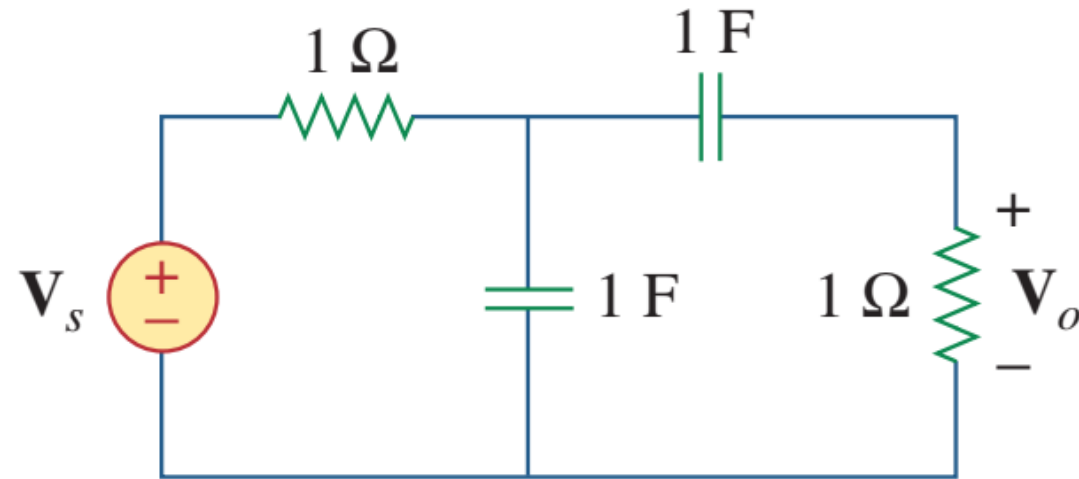
$$\left[-\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$

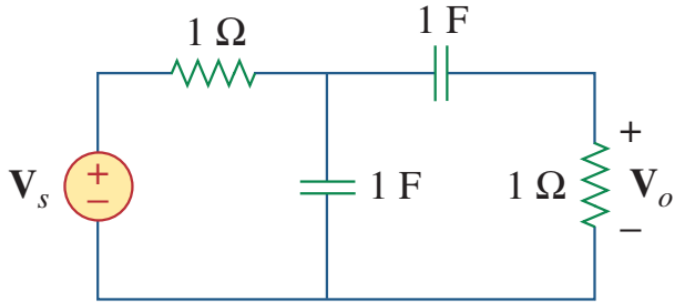
$$\left[\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right] \omega_0$$

Notes: (1) The expression for Q of the series RLC circuit is the inverse of that for Q of the parallel circuit. (2) For $Q \geq 10$, $\omega_1 \simeq \omega_0 - \frac{B}{2}$, and $\omega_2 \simeq \omega_0 + \frac{B}{2}$.

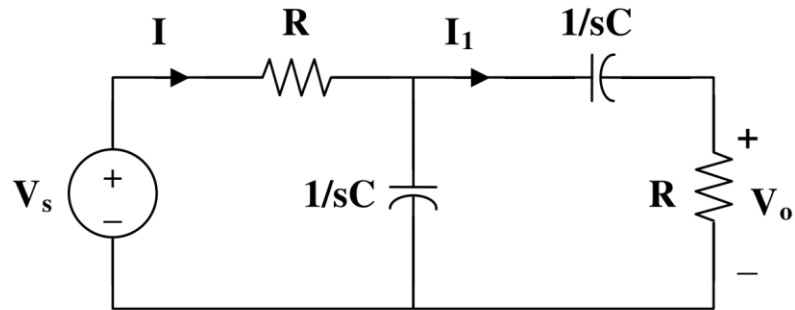
[Source: Berkeley]

2. Determine the center frequency and bandwidth of the bandpass filters in Fig. 14.88.





Consider the circuit below.



$$\mathbf{Z}(s) = R + \frac{1}{sC} \parallel \left(R + \frac{1}{sC} \right) = R + \frac{\frac{1}{sC} \left(R + \frac{1}{sC} \right)}{R + \frac{2}{sC}}$$

$$\mathbf{Z}(s) = R + \frac{1 + sRC}{sC(2 + sRC)}$$

$$\mathbf{Z}(s) = \frac{1 + 3sRC + s^2 R^2 C^2}{sC(2 + sRC)}$$

$$\mathbf{I} = \frac{\mathbf{V}_s}{\mathbf{Z}}$$

$$\mathbf{I}_1 = \frac{1/sC}{2/sC + R} \mathbf{I} = \frac{\mathbf{V}_s}{\mathbf{Z}(2 + sRC)}$$

$$\mathbf{V}_o = \mathbf{I}_1 R = \frac{R \mathbf{V}_s}{2 + sRC} \cdot \frac{sC(2 + sRC)}{1 + 3sRC + s^2 R^2 C^2}$$

$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{sRC}{1 + 3sRC + s^2 R^2 C^2}$$

$$\mathbf{H}(s) = \frac{1}{3} \left[\frac{\frac{3}{RC} s}{s^2 + \frac{3}{RC} s + \frac{1}{R^2 C^2}} \right]$$

$$\text{Thus, } \omega_0^2 = \frac{1}{R^2 C^2} \quad \text{or} \quad \omega_0 = \frac{1}{RC} = \mathbf{1 \text{ rad/s}}$$

$$B = \frac{3}{RC} = \mathbf{3 \text{ rad/s}}$$

3. A “general” first-order filter is shown in Fig. 14.93.

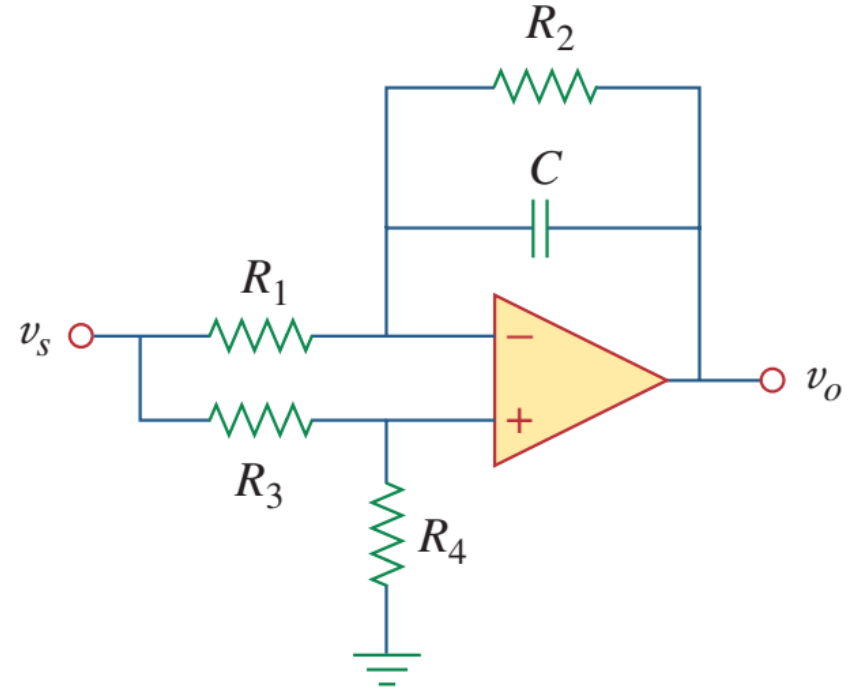
(a) Show that the transfer function is

$$\mathbf{H}(s) = \frac{R_4}{R_3 + R_4} \times \frac{s + (1/R_1 C)[R_1/R_2 - R_3/R_4]}{s + 1/R_2 C},$$

$s = j\omega$

(b) What condition must be satisfied for the circuit to operate as a highpass filter?

(c) What condition must be satisfied for the circuit to operate as a lowpass filter?



$$\begin{cases} \frac{V_1 - V_s}{R_1} + \frac{V_1 - V_o}{R_2} + \frac{V_1 - V_o}{\frac{1}{j\omega C}} = 0 & \textcircled{1} \\ \frac{V_1 - V_s}{R_3} + \frac{V_1}{R_4} = 0 & \textcircled{2} \end{cases}$$

From $\textcircled{2}$, we obtain $V_1 = \frac{R_4}{R_3 + R_4} V_s$ $\textcircled{3}$

From $\textcircled{1}$, $(R_1 + R_2 + R_1 R_2 j\omega C) V_1 - R_2 V_s - (R_1 + R_1 R_2 j\omega C) V_o = 0$ $\textcircled{4}$

Substitute $\textcircled{3}$ into $\textcircled{4}$,

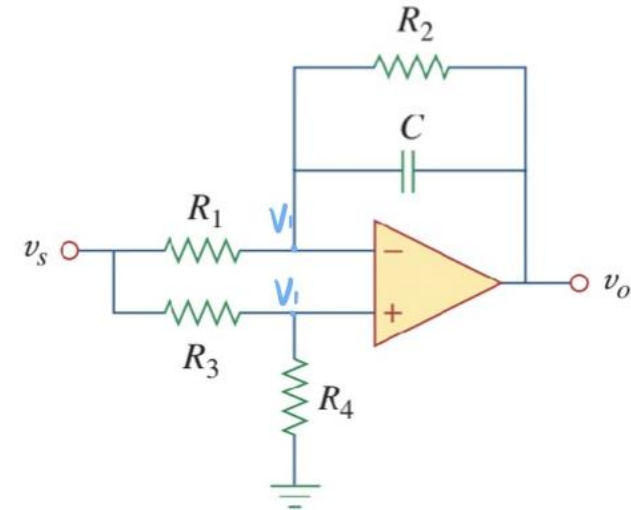
$$[(R_1 + R_2 + R_1 R_2 j\omega C) R_4 - R_2 (R_3 + R_4)] V_s = (R_1 + R_1 R_2 j\omega C) (R_3 + R_4) V_o$$

Simplify the above equation.

$$\frac{V_o}{V_s} = \frac{[(R_1 + R_2 + R_1 R_2 j\omega C) R_4 - R_2 (R_3 + R_4)]}{(R_1 + R_1 R_2 j\omega C) (R_3 + R_4)}$$

$$= \frac{R_4}{R_3 + R_4} \frac{\frac{1}{R_2 C} + j\omega - \frac{R_3}{R_4 R_1 C}}{\frac{1}{R_2 C} + j\omega}$$

$$= \frac{R_4}{R_3 + R_4} \frac{j\omega + \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right)}{\frac{1}{R_2 C} + j\omega} \quad s = j\omega$$



(b) When $R_1 R_4 = R_2 R_3$,

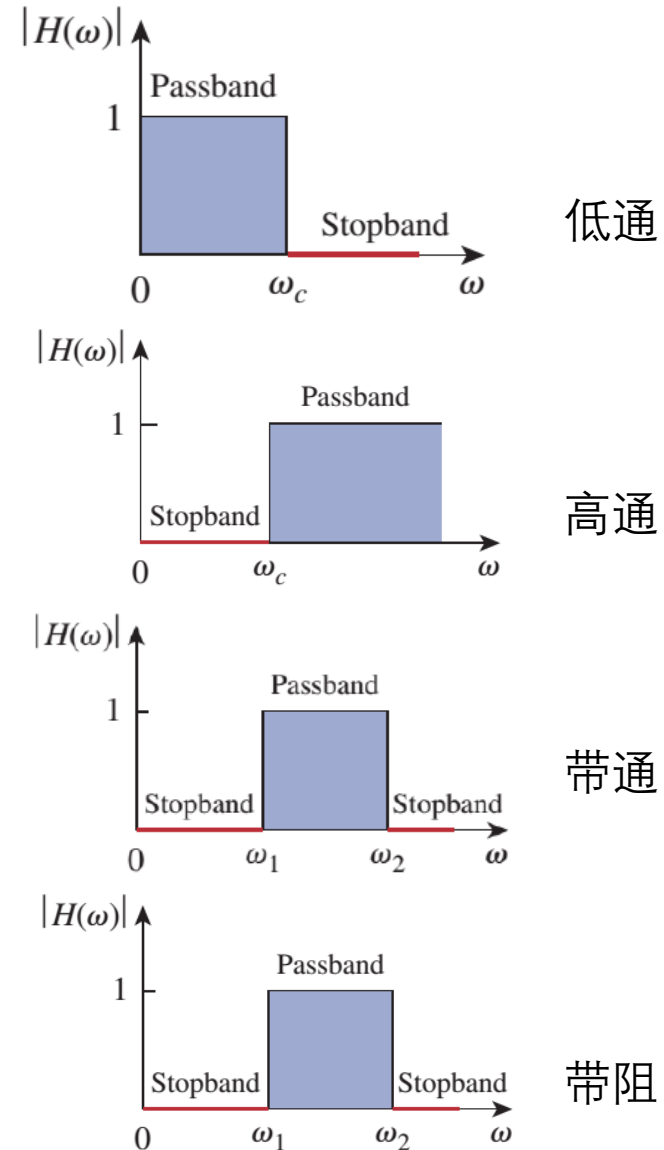
$$H(s) = \frac{R_4}{R_3 + R_4} \cdot \frac{s}{s + 1/R_2 C}$$

(c) When $R_3 \rightarrow \infty$,

$$H(s) = \frac{-1/R_1 C}{s + 1/R_2 C}$$

Filters

- Circuit designed to retain certain frequency range but discard or attenuate others
 - **Low-pass**: pass low frequencies and reject high frequencies
 - **High-pass**: pass high frequencies and reject low frequencies
 - **Band-pass**: pass some particular range of frequencies, reject other frequencies outside the band
 - **Band-reject (notch)**: reject a range of frequencies and pass all other frequencies

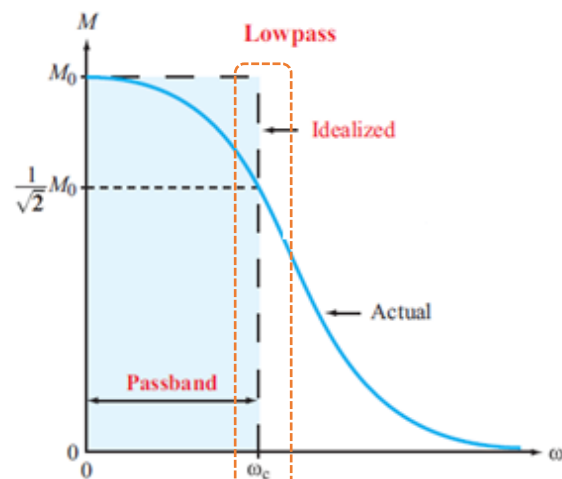


Filters — Realistic Curves

f_c : Cutoff frequency
(corner frequency)

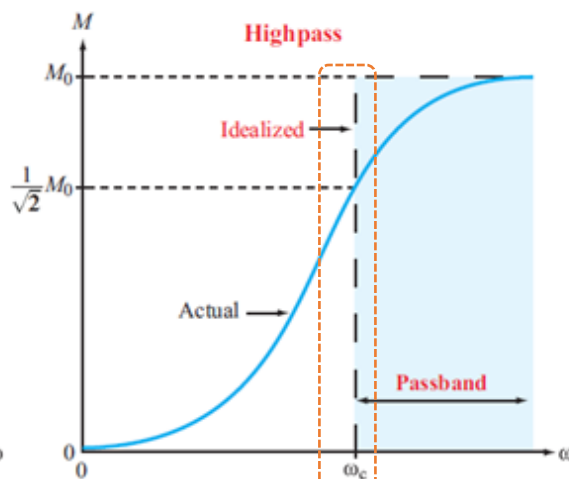
$$|H|_{f_c} = \frac{1}{\sqrt{2}} |H|_{\max}$$

低通



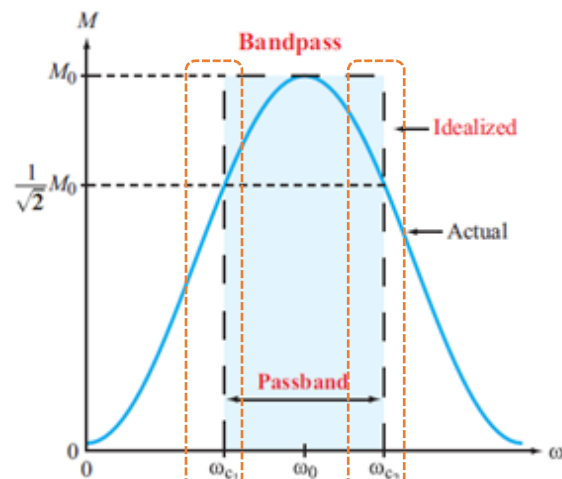
(a) Lowpass filter

高通



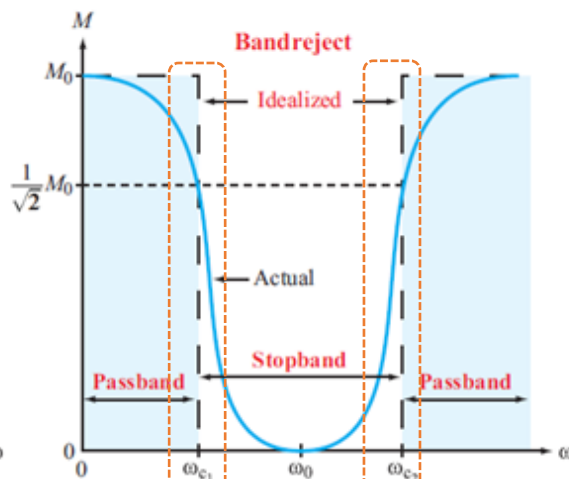
(b) Highpass filter

带通



(c) Bandpass filter

带阻

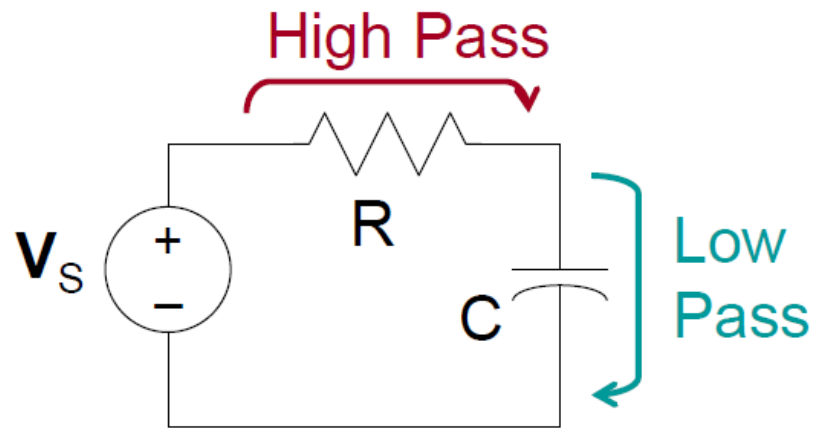


(d) Bandreject filter

Passive Filters

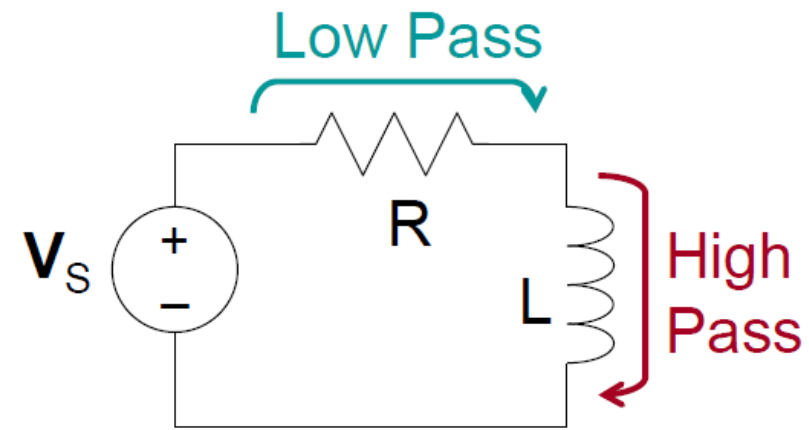
- A filter is passive if it only consists of passive elements
 - *R, L and C*
- LC filters have been used in practical applications for more than eight decades
 - Very important circuits
 - many technological advances would not have been possible without development of filters
 - LC filter technology feeds many areas
 - equalizers, impedance-matching networks, transformers, shaping networks, power dividers, attenuators, and directional couplers ...

First Order (RL / RC) Filter Circuits



$$H_R = R / (R + 1/j\omega C)$$

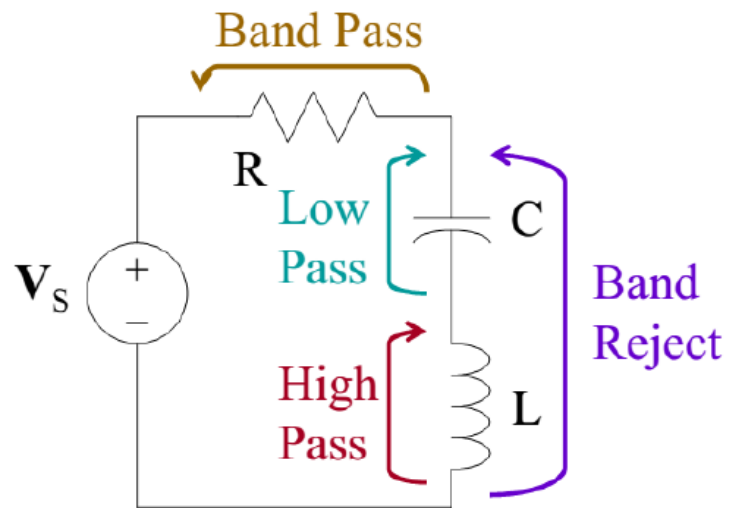
$$H_C = (1/j\omega C) / (R + 1/j\omega C)$$



$$H_R = R / (R + j\omega L)$$

$$H_L = j\omega L / (R + j\omega L)$$

Second Order (RLC) Filter Circuits



$$\mathbf{Z} = R + 1/j\omega C + j\omega L$$

$$\mathbf{H}_{\text{BP}} = R / \mathbf{Z}$$

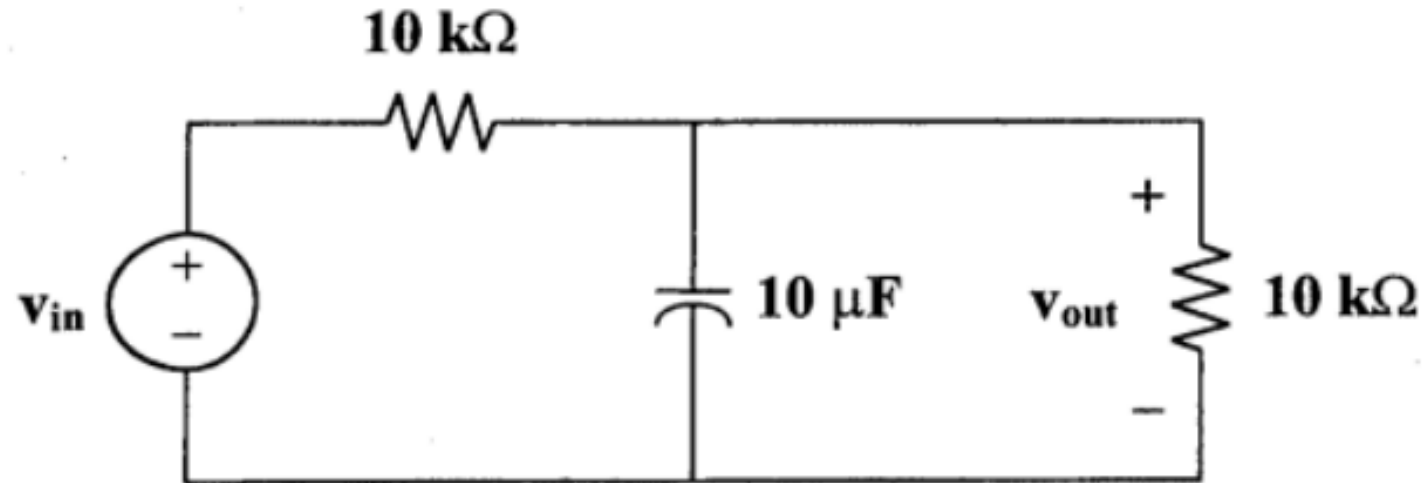
$$\mathbf{H}_{\text{LP}} = (1/j\omega C) / \mathbf{Z}$$

$$\mathbf{H}_{\text{HP}} = j\omega L / \mathbf{Z}$$

$$\mathbf{H}_{\text{BR}} = \mathbf{H}_{\text{LP}} + \mathbf{H}_{\text{HP}}$$

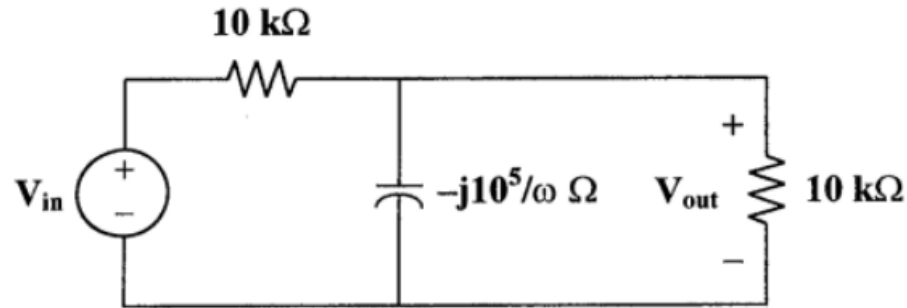
Example 1

What type of filter is represented by the circuit in the figure below?
What is the cutoff frequency(corner frequency)?



Example

In the frequency domain, the circuit is



Find the transfer function $H(\omega) = \frac{V_{out}}{V_{in}}$

$$= \frac{-j10^5/\omega || 10 \cdot 10^3}{-j10^5/\omega || 10 \cdot 10^3 + 10 \cdot 10^3}$$

$$= \frac{1}{2 + j\omega/10} \Rightarrow \text{low pass filter}$$

Since the voltage starts at $V_{out} = \frac{1}{2}V_{in}$ and as

$\omega \rightarrow \infty$ $V_{out} = 0$ we can look at a value for

$$|V_{out}| = \frac{0.707}{2} |V_{in}| = 0.3535 |V_{in}|$$

To find the frequency, find the value when

$$H(\omega) = 0.3505 = 1 / \sqrt{4 + \frac{\omega^2}{100}}$$

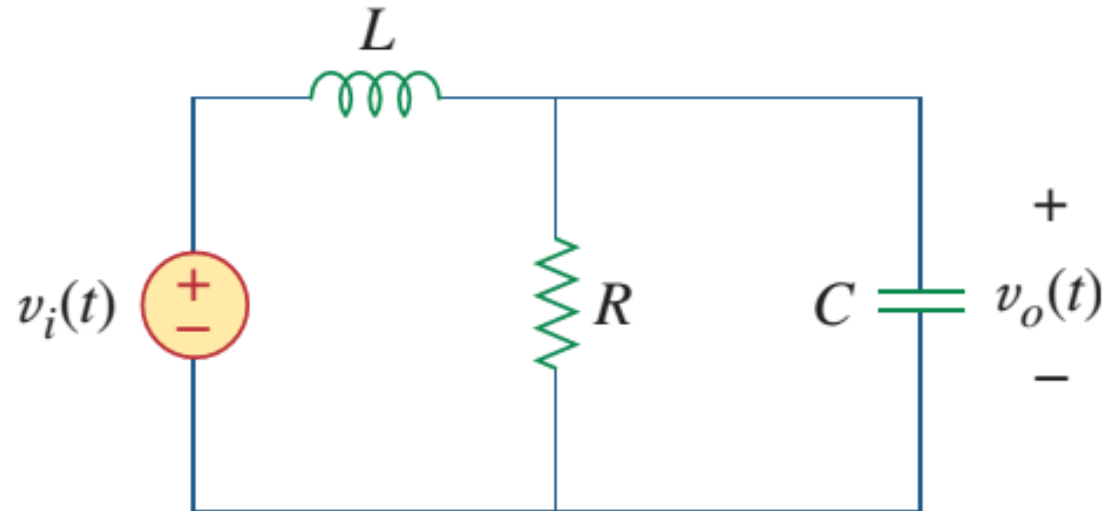
$$8 = 4 + \frac{\omega^2}{100} \Rightarrow \omega = 20\text{ rad/s}$$

Example 2

Determine what type of filter is shown below. Calculate half-power frequency.

$$R = 2k\Omega, L = 2H \text{ and } C = 2\mu F.$$

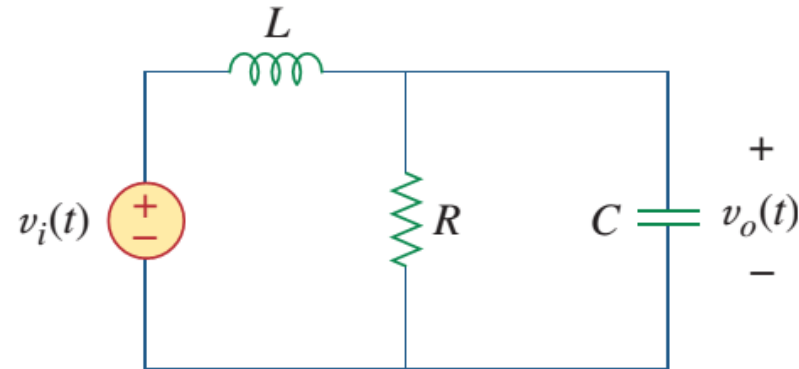
$$H(s) = \frac{V_o}{V_i} =$$



Example 2

Determine what type of filter is shown below. Calculate the cutoff frequency.

$$R = 2k\Omega, L = 2H \text{ and } C = 2\mu F.$$



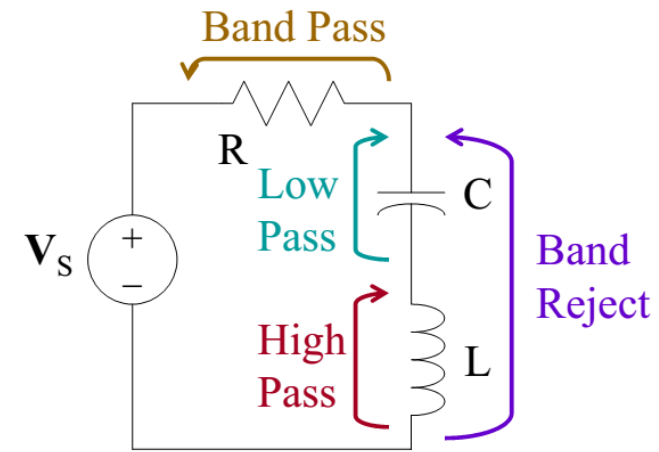
$$\begin{aligned} H(s) = \frac{V_o}{V_i} &= \frac{R // \frac{1}{j\omega C}}{j\omega L + R // \frac{1}{j\omega C}} = \frac{R}{R(1 - \omega^2 LC) + j\omega L} = \frac{R}{R(1 + s^2 LC) + sL} \\ &= \frac{250000}{s^2 + 250s + 250000} = \frac{250000}{(s + 125)^2 + 234375} \end{aligned}$$

Low-pass filter. $s = 0, H(s)_{max} = 1.$

$$\left| \frac{250000}{(s + 125)^2 + 234375} \right| = \frac{1}{\sqrt{2}} \rightarrow \omega = 35.049 \text{ Hz}$$

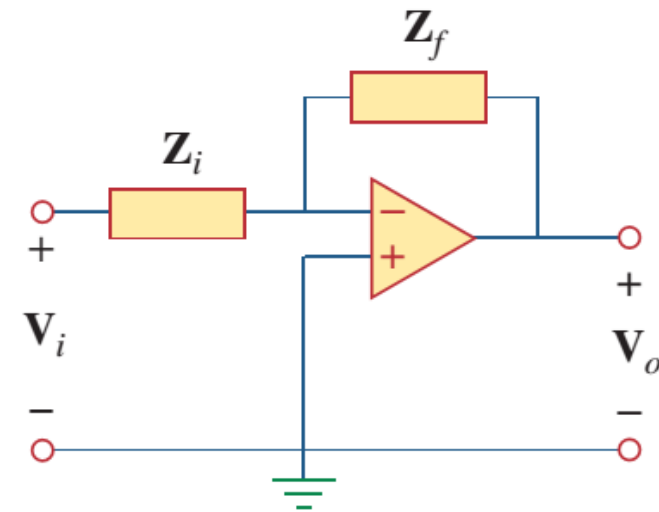
Active Filters

- Passive filters have a few drawbacks.
 - Generally, they cannot create gain greater than 1.
 - They do not work well for frequencies below the audio range (300Hz~3000Hz).
 - They require inductors, which tend to be bulky and more expensive than other components.



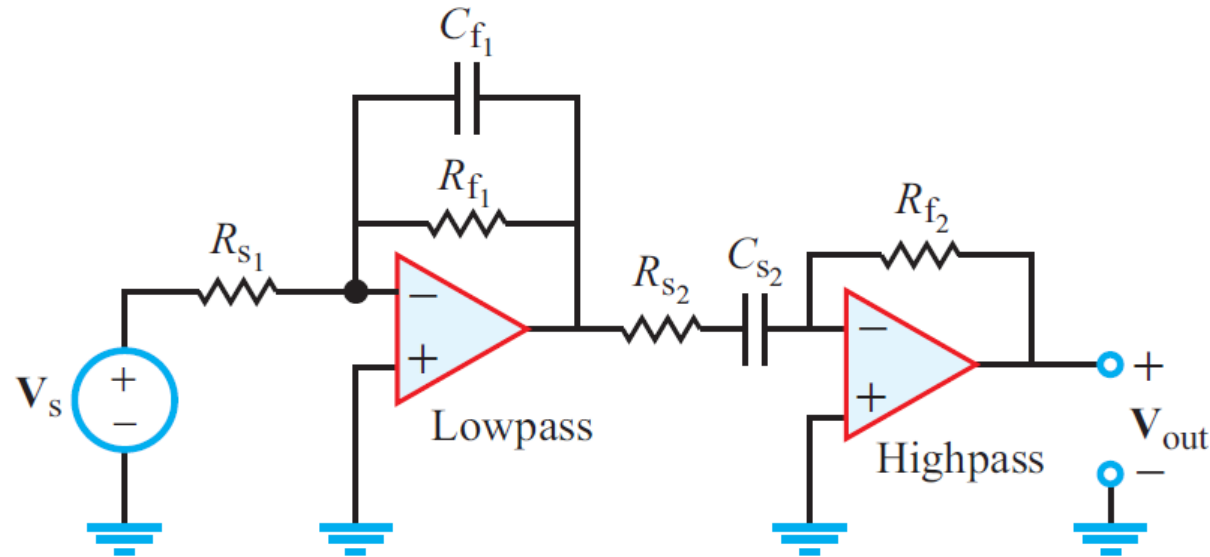
- It is possible, using op-amps, together with resistors and capacitors, to create all the common filters.

- Their ability to isolate input and output also makes them very desirable.
- Limited to frequency less than 1MHz.



Example 3: Bandpass Filter

The block diagram shown in Fig. 9-27(a) is a two-stage bandpass filter with the following elements: $R_{s1} = 1 \text{ k}\Omega$, $R_{f1} = 10 \text{ k}\Omega$, $C_{f1} = 9 \text{ nF}$, $R_{s2} = 12 \text{ k}\Omega$, $C_{s2} = 9 \text{ nF}$, and $R_{f2} = 96 \text{ k}\Omega$. Determine and plot the magnitude of the transfer function and obtain the values of ω_0 , ω_{c1} , ω_{c2} , B , and Q .



Solution: The first stage is a lowpass filter with a transfer function given by Eq. (9.88) as

$$\mathbf{H}_{\text{LP}}(\omega) = G_{\text{LP}} \left(\frac{1}{1 + j\omega/\omega_{\text{LP}}} \right)$$

with

$$G_{\text{LP}} = -\frac{R_{f1}}{R_{s1}} = -\frac{10^4}{10^3} = -10$$

and

$$\omega_{\text{LP}} = \frac{1}{R_{f1} C_{f1}} = \frac{1}{10^4 \times 9 \times 10^{-9}} = 11.11 \text{ krad/s.}$$

The transfer function of the highpass filter in the second stage is characterized by Eq. (9.91) as

$$\mathbf{H}_{\text{HP}}(\omega) = G_{\text{HP}} \left(\frac{j\omega/\omega_{\text{HP}}}{1 + j\omega/\omega_{\text{HP}}} \right)$$

with

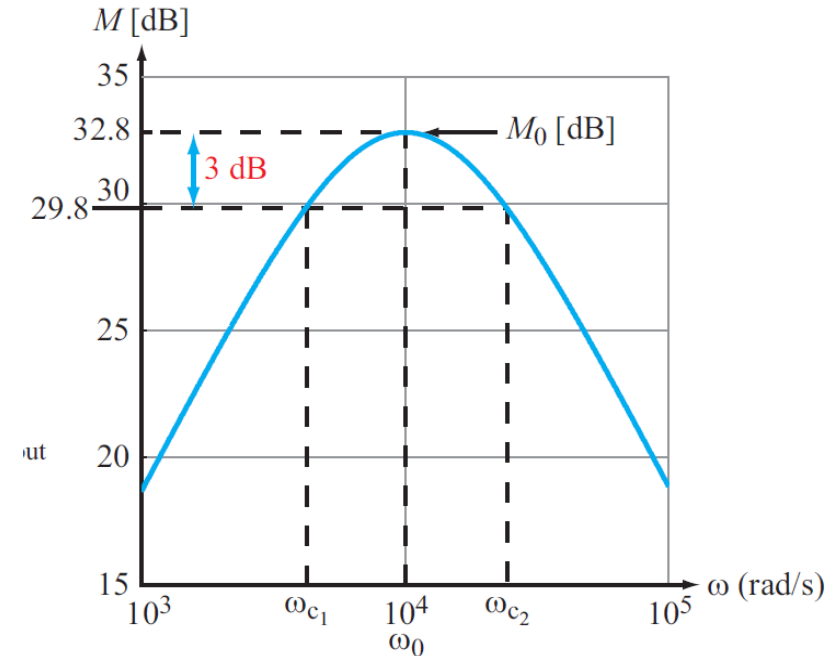
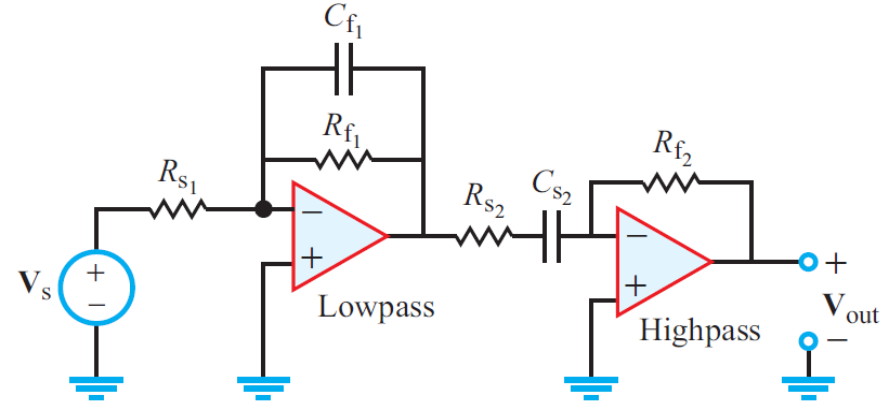
$$G_{\text{HP}} = -\frac{R_{f2}}{R_{s2}} = -\frac{96 \times 10^3}{12 \times 10^3} = -8,$$

and

$$\omega_{\text{HP}} = \frac{1}{R_{s2} C_{s2}} = \frac{1}{12 \times 10^3 \times 9 \times 10^{-9}} = 9.26 \text{ krad/s.}$$

The combined transfer function is then given by

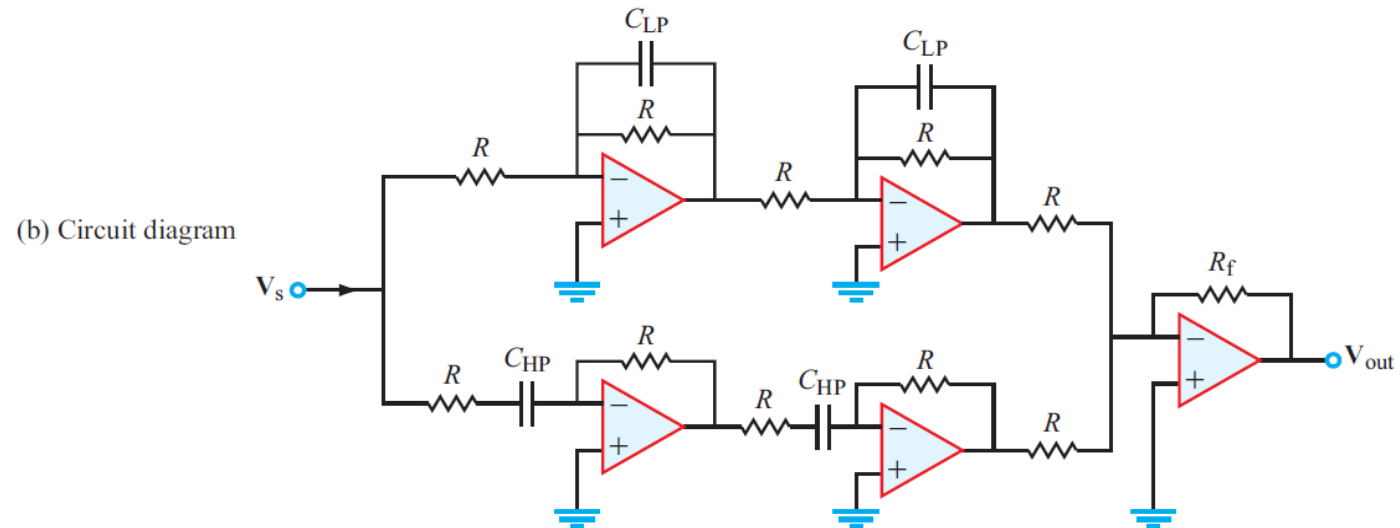
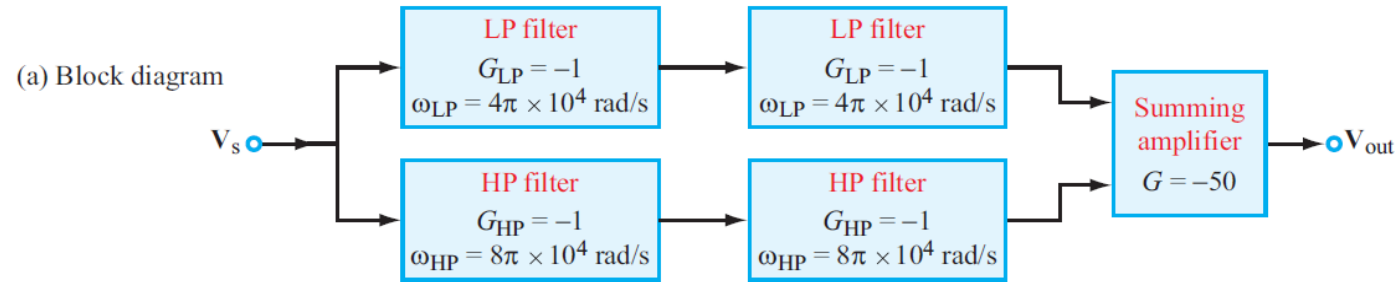
$$\begin{aligned} \mathbf{H}(\omega) &= \mathbf{H}_{\text{LP}} \mathbf{H}_{\text{HP}} \\ &= G_1 G_2 \left[\frac{j\omega/\omega_{\text{HP}}}{(1 + j\omega/\omega_{\text{LP}})(1 + j\omega/\omega_{\text{HP}})} \right], \quad (9.93) \end{aligned}$$



Example 4: Bandreject Filter

Design a bandreject filter with the specifications:

- (a) Gain = -50 , (b) bandstop extends from 20 kHz to 40 kHz, and (c) gain roll-off rate = -40 dB/decade along both boundaries of the bandstop.



Somewhat arbitrarily, we select $R = 1 \text{ k}\Omega$. From

$$\omega_{\text{LP}} = 4\pi \times 10^4 = \frac{1}{RC_{\text{LP}}} \rightarrow C_{\text{LP}} = 7.96 \text{ nF} \simeq 8 \text{ nF},$$

$$\omega_{\text{HP}} = 8\pi \times 10^4 = \frac{1}{RC_{\text{HP}}} \rightarrow C_{\text{HP}} \simeq 4 \text{ nF}.$$

The value of R_f is specified by the gain of the summing amplifier as

$$G = -50 = -\frac{R_f}{R} \rightarrow R_f = 50 \text{ k}\Omega.$$

The transfer function of the bandreject filter is given by

$$\begin{aligned} \mathbf{H}(\omega) &= G[\mathbf{H}_{\text{LP}}^2 + \mathbf{H}_{\text{HP}}^2] \\ &= -50 \left[\left(\frac{1}{1 + j\omega RC_{\text{LP}}} \right)^2 + \left(\frac{j\omega RC_{\text{HP}}}{1 + j\omega RC_{\text{HP}}} \right)^2 \right] \\ &= -50 \left[\left(\frac{1}{1 + j\omega/4\pi \times 10^4} \right)^2 + \left(\frac{j\omega/8\pi \times 10^4}{1 + j\omega/8\pi \times 10^4} \right)^2 \right]. \end{aligned}$$

(c) Filter frequency response

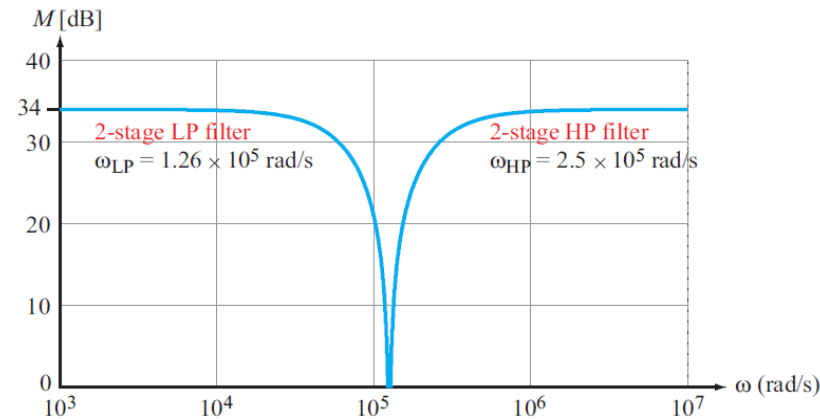


Figure 9-28: Bandreject filter of Example 9-12.

END